

The BB84 protocol

– Worksheet –

The BB84 protocol

1. Alice generates a random polarisation state.
 2. Bob measures the polarisation state using a random measurement basis.
 3. Alice and Bob record the basis they each chose and the polarisation state they generated / measured.
 4. Alice and Bob record the bit associated with the polarisation direction: 0 for h or $-$, 1 for v or $=$.
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5. Alice generates a random polarisation state.
 6. Bob measures the polarisation state using a random measurement basis.
 7. Alice and Bob record the basis they each chose and the polarisation state they generated / measured.
 8. Alice and Bob record the bit associated with the polarisation direction: 0 for h or $-$, 1 for v or $=$.

Important for the Advanced Experiments!

- For the colour signal decryption to work, the state counters of the communicating experiments must run in sync!
This means that the state counters must always show the same value *after* Bob's measurement.
- The Alice–Bob channel uses a different encoding than the Eve–Bob channel. In the Bob experiment, you must therefore first select *whose* signal is being received!

Experiment 1: Key generation

	Photon	Example	Example	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Alice	Basis	h/v	h/v	h/v																			
	State	v	v	v																			
	Bit	1	1	1																			
Bob	Basis	$+/-$	h/v	h/v																			
	State	$+$	v	v																			
	Bit	1	1	1																			
	Agreement of bases	$\times$	$\checkmark$																				
	Key	$/$	1																				

Experiment 2: Eavesdropping

	Photon	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Alice	Basis State Bit																				
Eve	Basis State Bit																				
Bob	Basis State Bit																				
	Agreement of bases (Alice/Bob)																				
	Alice's key																				
	Bob's key																				

Experiment 3: Covert key generation

	Photon	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Alice	Basis State Bit																				
Bob	Basis State Bit																				
	Agree- ment of bases																				
	Key																				

Experiment 4: Covert eavesdropping

	Photon	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Alice	Basis State Bit																				
Eve	Basis State Bit																				
Bob	Basis State Bit																				
	Agreement of bases (Alice/Bob)																				
	Alice's key																				
	Bob's key																				

Experiment 5: Eve, u there?

	Photon	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Alice	Basis																				
	State																				
	Bit																				
Bob	Basis																				
	State																				
	Bit																				
	Agreement of bases																				
	Alice's key																				
	Bob's key																				

